

Exposure to international trade lowers green voting and worsens environmental attitudes

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Charlotte Bez^{1,2,9}✉, Valentina Bosetti^{3,4,9}✉, Italo Colantone^{3,5,6,7,9} & Maurizio Zanardi^{8,9}

From a political perspective, advancing green agendas in democracies requires obtaining electoral support for parties and candidates proposing green platforms. It is therefore crucial to understand the factors driving green voting and attitudes. Yet, limited research has explored the role of economic determinants in this context. In this study we show that globalization, through the distributional consequences of import competition, is an important determinant of support for parties proposing green platforms. Our analysis covers the United States and 15 countries of Western Europe, over the period 2000–2019, with trade exposure measured at the level of subnational geographic areas. We find that higher trade exposure leads to lower support for more environmentalist parties and to more sceptical attitudes about climate change. Our empirical findings are in line with the theoretical channel of deprioritization of environmental concerns, as trade-induced economic distress raises the salience of economic issues.

Conserving the natural environment and mitigating the impacts of climate change represent critical contemporary challenges and are of paramount importance for policymakers on a global scale. From a political point of view, advancing green agendas in democracies requires public awareness and substantial electoral support for parties and candidates proposing environmental policies. Against this backdrop, it is crucial to reach a thorough understanding of the determinants of green voting and attitudes. In this respect, a large literature has looked into individual characteristics such as gender, age and education (for a review, see ref. 1). More recently, a growing strand of literature has also highlighted the role of personal experience of temperature abnormalities or extreme events^{2,3}. Yet, perhaps surprisingly, few studies have looked into the economic determinants of green voting and attitudes—focusing, for instance, on income and employment status⁴—and this has been typically done in terms of correlations rather than causal identification analysis.

In this study we set out to investigate the causal implications of changes in economic conditions on support for environmental action.

The guiding intuition is that situations of economic distress may lead to a deprioritization of environmental issues as economic concerns become more salient. In particular, economic shocks with distributional consequences, leading to unevenly distributed economic grievances, may have implications for the way people think about climate change and the extent to which they are willing to vote for parties that are more environmentalist.

As a source of variation in economic conditions, we focus on a key global trend, that of globalization. Specifically, we exploit differences in trade exposure across subnational geographic areas, as driven by historical differences in industry specialization. Although trade has generated aggregate welfare gains over the past three decades, it has also left losers behind. Rising trade exposure has been found to have strong distributional consequences, inducing economic distress in regions that were relatively more affected by import competition⁵. Our analysis covers the United States and 15 countries of Western Europe over the period 2000–2019. We find that higher trade exposure leads to lower support for more environmentalist parties and to

¹Scuola Superiore Sant'Anna, Pisa, Italy. ²Potsdam Institute for Climate Impact Research, Potsdam, Germany. ³Bocconi University, Milan, Italy. ⁴RFF-CMCC European Institute of Economics and the Environment, Euro-Mediterranean Center on Climate Change, Lecce, Italy. ⁵Baffi-Carefin Research Centre, Milan, Italy. ⁶CESifo, Munich, Germany. ⁷FEEM, Milan, Italy. ⁸University of Surrey, Guildford, UK. ⁹These authors contributed equally: Charlotte Bez, Valentina Bosetti, Italo Colantone, Maurizio Zanardi. ✉e-mail: c.bez@santannapisa.it; valentina.bosetti@unibocconi.it

more sceptical attitudes about climate change. We provide evidence consistent with the channel of deprioritization of environmental issues vis-à-vis economic concerns. Our findings suggest that addressing the distributional consequences of trade may be instrumental in making progress on climate action in the years to come.

From a theoretical point of view, different, and complementary, factors may contribute and compound to determine the link between trade exposure and environmentalism. First, in line with Inglehart's postmaterialism theory⁶, economic distress may foster the relevance of materialist values, such as job availability, and dampen the salience of postmaterialist values, such as environmentalism. A second and related psychological factor is based on the 'finite pool of worry' hypothesis⁷, which states that humans' cognitive resource constraints also imply finite emotional resources. Individuals exposed to the negative consequences of trade, and consequently worried about their economic and employment status, may reduce their concern about environmental threats. Third, environmental quality has been largely shown to be a normal good⁸, whose demand grows with income. To the extent that rising trade exposure determines a negative income shock, this may then translate into lower environmentalism. Fourth, rising inequality has been associated with lower aggregate demand for environmental policy⁹. To the extent that rising trade exposure increases inequality by generating winners and losers, this may further contribute to lowering environmentalism.

A fifth factor is associated with the common narrative that environmental policies are antithetical to economic growth and employment. Individuals who are already harmed by trade may then reject climate change to oppose climate policies and their associated costs, which may entail further distress for them. For instance, import shocks may mobilize workers in polluting industries against emission-reducing policies such as carbon taxes and cap-and-trade systems, which may reduce the competitiveness of domestic producers vis-à-vis foreign competitors operating in less-regulated countries. It has indeed been noticed how the potential losers of the green transition, for example, low-skilled manufacturing workers employed in polluting industries, tend to overlap significantly with social groups that have already been negatively affected by globalization and technological change over the past three decades¹⁰.

A sixth factor is more political in nature. A growing stream of literature has started to investigate the political implications of globalization. In particular, trade exposure has been shown to raise support for protectionist, isolationist and nationalist parties and candidates^{11,12}. It has also tilted people's attitudes in a nativist and authoritarian direction, thus fuelling support for radical-right parties¹³. These political forces tend to be also climate sceptical and hostile to policies prescribing climate change mitigation¹⁴. In particular, in their populist narrative, they tend to portray climate issues as concerns of the elites. These elites do not have to worry about the more serious economic problems faced by the common people and may push for green policies that disproportionately hit lower-income people¹⁵. As trade-exposed voters identify more with these parties, their attitudes may also be tilted in the same direction through party cues, thus reinforcing the deprioritization of environmental issues¹⁶.

Although we provide evidence consistent with the key theoretical channel of deprioritization of environmental concerns, it is beyond the scope of this paper to assess the relative importance of the different theoretical factors underlying this process. Such relative roles may be hard to identify even in experimental settings, where one can artificially induce one shock at a time, given the similarity of the factors and the psychological complexity of attitude formation¹⁷.

In this study we investigate the environmental dimension of trade-induced political shifts, which has to date been little explored. Indeed, while international trade is often discussed because of its impact on the environment, we do not have evidence on its possible direct impact on support for environmental action. Yet, shedding light

Table 1 | Baseline effects of trade exposure on voting at the regional level

	(1)	(2)	(3)	(4)
Dependent variable	Environmentalism			
Sample	United States		Europe	
Trade exposure	-0.138*** (0.008)	-0.037*** (0.008)	-0.049*** (0.007)	-0.080*** (0.020)
Estimator	OLS	2SLS	OLS	2SLS
Commuting zone effects	Yes	Yes	–	–
Commuting zone—specific trends	Yes	Yes	–	–
Region effects	–	–	Yes	Yes
Region-specific trends	–	–	Yes	Yes
Observations	7,241	7,241	9,634	9,634
R ²	0.54	-	0.38	-
First-stage results				
World export supply	–	0.058*** (0.001)	–	0.006*** (0.001)
Kleibergen–Paap F-statistic	–	9,375.1	–	26.51

Regression coefficients refer to the effect of a 1 s.d. change in trade exposure. Standard errors in parentheses. In columns 1 and 2, standard errors are clustered by commuting zone. In columns 3 and 4, standard errors are clustered by region–year. *** $P < 0.01$, ** $P < 0.05$, * $P < 0.1$.

on this issue is key to improving our understanding of how to advance green agendas in Western democracies.

The empirical analysis is organized in three steps. First, we investigate the impact of trade exposure on voting at the subnational level. Second, we replicate the voting analysis using data at the individual level. Finally, we assess the effect of trade exposure on individual attitudes. Throughout the analysis, the main explanatory variable of interest is a measure of exposure to import competition in the years prior to the elections or interviews. This is computed at the level of subnational geographic areas. The methodological details are described below in Methods. Intuitively, this measure is meant to capture economic distress induced by trade on the 'supply side'. In the baseline analysis we consider exposure to imports from all countries, but we also show results referring specifically to imports from high-income countries, low-income countries and China, as these may have heterogeneous implications across different regions. We address the potential endogeneity of imports by following a standard approach in international economics. That is, the growth in imports from a given group of trading partners is instrumented using the growth in exports by the same trading partners to other destination countries.

Regional-level results on voting

The main dependent variable in the regressions on voting at the level of subnational geographic areas is the Environmentalism Score. This index is the average of the environmentalism scores of all the parties competing in each election, weighted by their vote shares in each area. Intuitively, it is a proxy for the ideological leaning of subnational geographic areas in terms of environmentalism, as inferred from electoral outcomes in each election. Descriptive evidence on the dynamics of environmentalism scores and other variables of interest is presented in the Supplementary Information.

United States

Results for the US analysis are reported in the first two columns of Table 1. This analysis is conducted at the level of commuting zones, based on House elections. The estimated coefficients on trade exposure

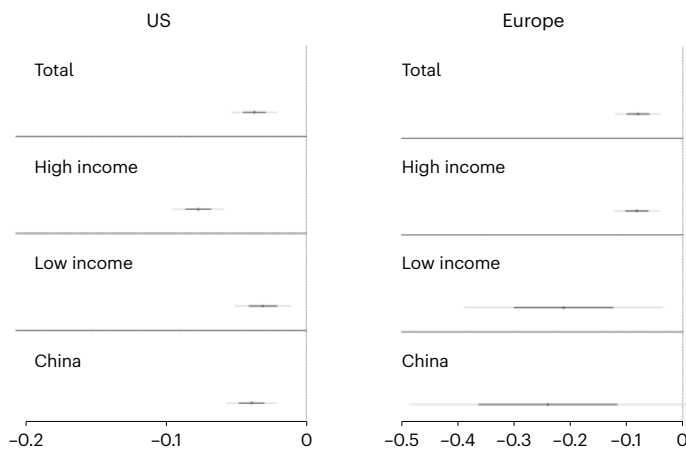


Fig. 1 | Effects of trade exposure from different origins. Instrumental variable estimates of the impact of a 1 s.d. change in trade exposure on environmentalism in the United States (left) and in Europe (right). Lines around the point estimates show 90% and 95% confidence intervals. Full results are shown in Supplementary Tables 4 and 19.

can be interpreted as the effect of a 1 s.d. change. The first column reports the ordinary least squares (OLS) results, while the second refers to the two-stage least squares (2SLS) regression. The estimated effect of trade exposure is negative and statistically significant, suggesting that growing import competition reduces support for more environmentalist political forces. As for the magnitude of this effect, a 1 s.d. increase in total trade exposure reduces the environmentalism score by around 4% of a standard deviation (see the instrumental variable estimate in column 2). The results considering imports from different origins are fairly stable across the board (Fig. 1, left panel).

Europe

For Europe, we conduct the analysis at the level of electoral districts, based on national legislature elections. Results are reported in columns 3 and 4 of Table 1. Consistent with the US results, the estimated effect of trade exposure is negative and statistically significant. As for the magnitude, a 1 s.d. increase in total trade exposure reduces the environmentalism score by around 22% of a standard deviation (see the instrumental variable estimate in column 4). The estimated effects are not statistically different from each other when we consider different origins of imports (Fig. 1, right panel).

Robustness and extensions

We have carried out a large number of robustness checks and extensions to the regional analysis of electoral outcomes. These are listed fully in Methods and are reported with all details in the Supplementary Information. In particular, for both the United States and Europe, results are robust to: excluding the years of the financial crisis; controlling for export exposure; controlling for ideological leaning in terms of protectionism; controlling for macroeconomic conditions; controlling for exposure to extreme weather events, that is, temperature anomalies, heat episodes and dry spells, as in ref. 3; and controlling for other potential confounders.

In addition, for the United States we also show that our evidence is robust to: using alternative environmentalism scores; controlling for the vote share of the Democratic party in the first presample election, interacted with either a linear time trend or year fixed effects; and considering both House and Senate elections at the level of analysis at which they are held, that is, congressional-district and state level, respectively. For Europe, we obtain similar results as the baseline when: using as a dependent variable the cumulative share of district votes for parties identified as being green; dropping the largest regions within

each country and overall; replicating the analysis at the more disaggregated NUTS-3 level (nomenclature of territorial units for statistics to identify small regions, available for Italy and the United Kingdom); and excluding countries with a majoritarian or mixed electoral system.

Individual-level results on voting and attitudes

We accompany the analysis of voting at the level of subnational geographic areas with an analysis based on individual-level survey data. First, we replicate the analysis of voting using information on the party voted by each individual in a given election. This allows us to investigate potential heterogeneity in the effect of trade exposure across different categories of individuals. Second, we complement the analysis of voting with an investigation of the impact of trade exposure on green attitudes.

Voting in the United States

The first two columns of Table 2 display the results for individual voting in the United States. The dependent variable is measured based on the party voted by each individual in a given election. The estimated effect of trade exposure is negative and statistically significant, with a magnitude very close to that found in the commuting-zone level analysis. Specifically, a 1 s.d. increase in trade exposure reduces the environmentalism score by about 3% of a standard deviation (see the instrumental variable estimate in column 2). In Supplementary Table 18 we show that this effect is stable when controlling for the self-reported partisan ideology of each individual, that is, Democratic, Republican, independent, or other.

This analysis at the individual level corroborates the subnational-level findings; moreover, it makes it possible to investigate potential heterogeneity of the effect. To this purpose, we have augmented the baseline instrumental variable regression with interactions between trade exposure and several individual characteristics (the results are shown in Supplementary Table 17). These include gender, education, type of occupation (that is, part time or full time), employment status (that is, unemployed, retired or student) and age groups (that is, below 25 or above 64). The evidence points to a ‘sociotropic’ response of voters, who seem to react to import competition independently from their personal extent of exposure, which may vary along with their individual characteristics. Only part-time workers seem to be non-responsive, with an overall estimated effect of trade exposure that is far from statistically significant. Instead, students and retired people, if anything, seem to show a stronger than average response to import competition hitting their commuting zone of residence.

Voting in Europe

The estimated effect of trade exposure is negative and statistically significant also in the analysis of Europe (columns 3 and 4 of Table 2). As for the magnitude, a 1 s.d. increase in trade exposure reduces the environmentalism score by around 15% of a standard deviation. This effect remains essentially unaffected when excluding from the sample those individuals who report voting for a radical-right party, according to the list given in ref. 18 (Supplementary Table 32). The heterogeneity analysis, similar to the US one, is presented in Supplementary Table 31. The estimated overall effect of trade exposure is statistically significant for all groups of individuals, pointing again to a ‘sociotropic’ response of voters. At the same time, there is some evidence of heterogeneous effects. In particular, the effect of import competition seems to be significantly stronger for unemployed individuals, and milder for females, white-collar workers and more educated individuals, who tend to be relatively sheltered from the economic impact of trade exposure in advanced economies.

Attitudes

We complement the analysis of voting with an investigation of the impact of trade exposure on green attitudes. This analysis is particularly important as it allows us to explore whether the trade-induced shift

Table 2 | Baseline effects of trade exposure on voting at the individual level

	(1)	(2)	(3)	(4)
Dependent variable	Individual environmentalism			
Sample	United States	Europe		
Trade exposure	-1.629*** (0.387)	-1.143** (0.468)	-0.021* (0.011)	-0.201*** (0.048)
Estimator	OLS	2SLS	OLS	2SLS
Female, age, education	Yes	Yes	Yes	Yes
Commuting zone effects	Yes	Yes	–	–
Year effects	Yes	Yes	–	–
Region effects	–	–	Yes	Yes
Region-specific trends	–	–	Yes	Yes
Observations	221,012	221,012	116,445	116,445
R ²	0.13	–	0.35	–
First-stage results				
World export supply	–	0.059*** (0.002)	–	0.002*** (0.000)
Kleibergen–Paap F-statistic	–	855.35	–	26.58

Regression coefficients refer to the effect of a 1 s.d. change in trade exposure. Standard errors in parentheses. In columns 1 and 2, standard errors are clustered by commuting zone–year. In columns 3 and 4, standard errors are clustered by region–year. *** $P < 0.01$, ** $P < 0.05$, * $P < 0.1$.

in voting away from environmentalist political forces is also related to a consistent shift in individual attitudes in the same direction. Specifically, we begin by focusing on climate change attitudes. Then, we consider additional survey items that allow us to provide evidence in line with the theoretical channel of deprioritization of environmental issues vis-à-vis economic concerns, as driven by trade exposure.

All the coefficients in Fig. 2 are obtained from instrumental variable regressions; they refer to total trade exposure and can be interpreted as the effect of a 1 s.d. change. For the United States, all the estimated effects are negative and statistically significant, based on the same specification employed for the analysis of individual vote (Fig. 2, left panel). Individuals living in commuting zones more exposed to import competition are less likely to think that the seriousness of climate change is correctly assessed or underestimated by the news ('Serious'), and less likely to worry about climate change ('Worry'). In terms of magnitudes, a 1 s.d. increase in trade exposure leads to a decrease by about 3 percentage points in the probability of perceiving climate change as a serious threat or being worried about it. Consistent with the channel of deprioritization of environmental concerns, individuals more exposed to trade are also: (1) less likely to consider themselves as being environmentalists; (2) less likely to mention environment/pollution/climate change as the most important problem; and (3) less likely to think that environmental protection is much (or somewhat) more important than job availability. In addition, more trade-exposed individuals are also less likely to support the statement that states should be required to use a minimum amount of renewable fuels, less likely to be employed, and less likely to report a household income in the top 10% of the distribution (the results are shown in Supplementary Table 35).

Similarly, in Europe, individuals living in regions more exposed to trade are significantly less likely to think of climate change as a serious issue (Fig. 2, right). Specifically, a 1 s.d. increase in total trade exposure reduces the probability of perceiving climate change as a serious problem by about 2 percentage points. More trade-exposed individuals are also: (1) less likely to think that fighting climate change and using energy more efficiently can boost the economy and jobs; (2) more likely to state that their household income is insufficient to have a minimum

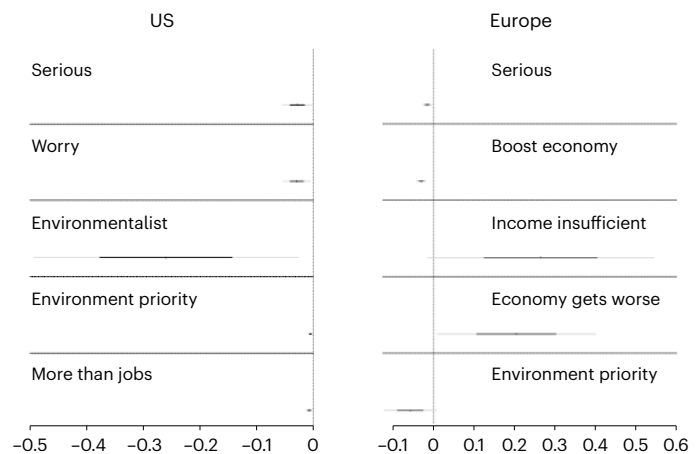


Fig. 2 | Effects of trade exposure on attitudes. Instrumental variable estimates of the impact of a 1 s.d. change in trade exposure on attitudes in the United States (left) and in Europe (right). Lines around the point estimates show 90% and 95% confidence intervals. Full results are shown in Supplementary Tables 33–35 and 38–40.

acceptable standard of living; (3) more likely to expect the economic situation of their country to get worse over the next year; and (4) less likely to agree with the statement that protecting the environment should be a priority, even if it affects economic growth.

Several robustness checks and extensions are presented in Methods, and reported with all details in the Supplementary Information, focusing on four main items of interest (that is, the two items on top of Fig. 2, for both the United States and Europe). In particular, results are largely robust when we consider exposure to imports from different origins and when we control for exposure to extreme weather events. We have also performed the same heterogeneity analysis as for voting. By and large, results point to a 'sociotropic' response of individuals, consistent with the findings on voting.

Overall, the evidence presented in this section supports the theoretical argument that trade exposure can have a negative impact on environmentalist voting by leading to the deprioritization of environmental issues amid rising economic concerns.

Conclusion

In this study we investigate the impact of globalization on green voting and attitudes. We find that higher trade exposure leads to lower support for environmentalist parties and worse attitudes about climate change. Our results suggest that mitigating the unequal repercussions of international trade is key to advancing support for green agendas in Western democracies. Our findings also bring to the forefront an important additional element for the evaluation of a currently debated policy: the carbon border adjustment mechanism. By internalizing the climate externality, tariff adjustments might lower import growth and partly avoid further deterioration of the socioeconomic situation of trade-exposed social groups. In addition, this mechanism would generate tariff revenues that, together with revenues from emissions-trading schemes such as the EU ETS, could be used to compensate vulnerable households.

Although we focus on international trade as a driver of politically salient distributional consequences, future studies may exploit other identifiable economic shocks such as automation, which is another key dimension of structural economic change. More research connecting economic factors, inequality and support for climate action is much needed, especially because transitioning to a low-carbon economy may itself imply reinforced/additional distributional consequences. The climate challenge will be much harder to win without a more inclusive policy approach to managing structural changes.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41558-023-01789-z>.

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Methods

The main explanatory variable of interest is a measure of exposure to import competition in the years prior to the elections or interviews. This is computed at the level of subnational geographic areas, following the theory-based approach developed by⁵:

$$\text{Trade Exposure}_{crt} = \sum_j \frac{L_{rj}(\text{presample})}{L_r(\text{presample})} \times \frac{\Delta \text{IMP}_{cjt}}{L_{cj}(\text{presample})} \quad (1)$$

where c indexes countries, r subnational geographic areas, j manufacturing industries and t years. ΔIMP_{cjt} is the change in (real) imports over the past n years, in country c and industry j . In the baseline analysis, in line with ref. 11, we consider the change in imports over the past two years. This is normalized over $L_{cj}(\text{presample})$: the number of workers in country c and industry j , measured presample and kept constant throughout the analysis. To obtain the subnational measure of trade exposure, we take the weighted sum of the change in imports per worker across industries, where the weights reflect the relative importance of the different industries in a given area, evaluated presample and kept constant throughout the analysis. Using presample employment figures is meant to avoid contamination issues stemming from the fact that current employment may be affected by import dynamics. Our measure is in fact meant to exploit variation in the ex ante vulnerability to trade across geographic areas, based on their historical employment composition. The idea is that different areas are more or less exposed to import pressure depending on their initial industrial structure. In particular, larger shocks are assigned to areas in which manufacturing was more relevant ex ante. However, given the same manufacturing share, larger shocks are attributed to areas historically specialized in industries witnessing stronger growth in imports over the sample, and in years in which the rise in imports is higher.

Our sample includes the United States and 15 countries of Western Europe: Austria, Belgium, Finland, France, Germany, Greece, Ireland, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom. Trade data are sourced at the product level from UN Comtrade-BACI, and aggregated at the level of NACE-subsection industries. Subsections are identified by two-character alphabetical codes (from DA to DN in rev. 1.1). They correspond to two-digit industries or aggregations of them (Supplementary Table 1). For the United States, employment data come from County Business Patterns and refer to the year 1988. For European countries, we obtain employment data either from Eurostat or from national sources. Depending on the country, the data refer to an initial year between 1988 and 1995 (see Supplementary Table 2 for details). For the United States, in the main analysis based on House elections, trade exposure is computed at the commuting-zone level; in the extensions, it is computed either at the state level—for the analysis of Senate elections—or at the congressional-district level—for the analysis of House elections. For European countries, in the main analysis we focus on NUTS-2 administrative regions, as in ref. 11. NUTS-2 regions have populations ranging between 800,000 and 3 million. In total, we have 192 regions in the sample. For Germany, employment figures are only available at the more aggregated NUTS-1 level. This implies that 16 out of 192 sample regions correspond to NUTS-1 regions. For an extension of the analysis, available for Italy and the United Kingdom, we measure trade exposure at the more disaggregated NUTS-3 level.

In the baseline analysis, we regress electoral outcomes and individual attitudes on total trade exposure, instrumented as follows:

$$\text{Instrument}_{crt} = \sum_j \frac{L_{rj}(\text{presample})}{L_r(\text{presample})} \times \frac{\Delta \text{WES}_{jt}}{L_{cj}(\text{presample})} \quad (2)$$

where ΔWES_{jt} is the growth of trading partners' exports in industry j to the rest of the world (that is, excluding country c), while the normalization and weighting are the same as in equation (1). Intuitively,

we instrument the growth in imports in a given sample country using the growth in exports by all its trading partners to all other countries. This instrument borrows from earlier work in international economics (for example, ref. 19). It is meant to capture the variation in imports that is due to exogenous changes in supply conditions abroad, rather than to domestic factors potentially correlated with attitudes and voting.

We show results based on four different measures of trade exposure. They are all computed according to equation (1), but consider different types of import flows in the ΔIMP_{cjt} term. Specifically, the baseline measure includes total imports from all trading partners, thus providing the most comprehensive account of import competition. The second measure focuses only on imports from high-income countries (that is, all the countries that have been classified by the World Bank as high income in at least one year over the period 1995–2019). The third measure considers imports from 52 low-income countries (as in ref. 20; the full list is given in Supplementary Table 3), while the fourth measure focuses on imports from China only. The underlying intuition is that imports from different sources may be more or less relevant for different industries and potentially have heterogeneous implications for different regions. Focusing on China actually fits our instrumental variable approach particularly well. In fact, as discussed by ref. 21, China is a trading partner witnessing a rapid and substantial internal transformation over the time period that we consider, thus providing clear supply shocks that we exploit for identification.

When using each alternative measure of trade exposure, the instrumental variable is also adapted to consider only exports originating from the selected group of trading partners. For instance, when focusing on Chinese imports, for the United States we follow closely ref. 5, using Chinese exports to a group of eight high-income countries: Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain and Switzerland. Similarly, for Europe we consider Chinese exports to the same destinations as in ref. 11. Specifically, compared to the eight export destinations employed by ref. 5, ref. 11 dropped the European countries, while adding Canada. Hence the final group contains: Australia, Canada, Japan and New Zealand. The same approach is taken for instrumenting imports from high- and low-income countries.

Voting data

For the United States, election data at the subnational level are from D. Leip's US Election Atlas (<https://uselectionatlas.org/>) and the MIT Election Data and Science Lab (<https://electionlab.mit.edu/>). For Europe, they are obtained from the Constituency-Level Elections Archive (CLEA²²) and a number of national sources. Our sample includes legislative elections taking place between 2000 and 2019. Our main variable of interest is a proxy for the ideological leaning of subnational geographic areas in terms of environmentalism. This is computed by combining party vote shares in each area with a party-specific indicator of environmentalism.

Specifically, we compute the centre of gravity of environmentalism in each area through the following formula:

$$\text{Environmentalism}_{st} = \frac{\sum_{p=1}^n w_{pst} \times \log(0.5 + \text{Score}_{pt})}{\sum_{p=1}^n w_{pst}} \quad (3)$$

where s indexes subnational geographic areas, p parties and t years (elections). w_{pst} is the area-specific vote share and Score_{pt} is the party-specific environmentalism score. Intuitively, the centre of gravity ($\text{Environmentalism}_{st}$) is the average of the environmentalism scores of the competing parties, weighted by their vote shares in the area. For brevity, in the paper we refer to the centre of gravity simply as the environmentalism score.

For the main analysis of the United States, based on House elections, we compute the environmentalism score at the level of commuting zones. As weights, we employ the commuting-zone-specific

two-party vote shares of the Democratic and Republican parties. These are computed as in ref. 12, taking the ratio of Democratic (or Republican) votes over the sum of votes cast for the Democratic and Republican parties. Reflecting the well-known heterogeneity of party platforms across states²³, the environmentalism score of each party is state-specific. It is computed as the average roll-call environmentalism score of senators from the relevant state over the past two years before the election, as obtained from the League of Conservation Voters. For extensions of the analysis, we compute the environmentalism score either at the congressional district level, based on party vote shares in the House elections, or at the state level, based on party vote shares in the Senate elections. In both cases, we keep using the same state-specific environmentalism scores based on League of Conservation Voters data. These scores are also employed in the analysis of voting at the individual level, which is based on survey data from the Cooperative Congressional Election Study (CCES) spanning the period 2006–2018.

For the 15 European countries in our sample, we use as weights the district-specific party vote shares for the elections of the legislative lower house. As for the party-specific environmentalism scores, we employ the environmental protection index made available by the Manifesto Project (MP²⁴). The MP provides human coding of statements made in party manifestos, making it possible to compute measures of the ideological leaning of parties along several dimensions (see ref. 11 for an analysis of nationalism). The environmental protection index (MP item per501) provides information on the number of claims in the party manifestos in favour of protecting the environment, fighting climate change and other ‘green’ policies. Being related to the national manifestos, party environmentalism scores are nation-election specific. These scores are also used in the individual-level analysis, which is based on survey data from the European Social Survey, with interviews spanning the period 2002–2019.

We note that the environmentalism scores employed for the United States and Europe are somewhat different. In fact, while for Europe they are based on party manifestos, for the United States they are based on roll-call data, due to the lack of information on individual candidates’ manifestos. In Supplementary Table 13 we show that the results for the United States are robust to using the MP environmentalism scores for the Democratic and Republican parties, thus following exactly the same approach as for Europe. For the United States, however, we think this approach is suboptimal since these scores are only available for Presidential elections. That is, there are no party manifestos coded for ‘stand-alone’ House and Senate elections. Moreover, being related to Presidential elections, MP scores do not vary by state, while we know that parties’ leanings may vary significantly across states.

For an extension of the analysis of European countries we also consider a second dependent variable at the district level: Green Share. This is the cumulative share of district votes for parties identified as being green. Specifically, these are all the parties categorized as ecological by the MP. In a further refinement, we also distinguish, within the group of green parties, those that have a primarily domestic focus from those that have a more general approach. This assessment is based on information we retrieved for all green parties from the party websites. The list of parties belonging to each of the two groups is provided in Supplementary Information, section 2.

Attitudes data

For the United States, we rely on data from the environment survey of the Gallup Poll Social Series and from the CCES. For Gallup, data span the period 2000–2019 and cover a representative sample of around 1,000 respondents per year. We employ four survey items:

- ‘Thinking about what is said in the news, in your view is the seriousness of global warming generally exaggerated, generally correct, or is it generally underestimated?’ From this question,

we obtain a dummy equal to one in case of ‘generally correct’ or ‘underestimated’, and zero otherwise.

- ‘How much do you personally worry about the greenhouse effect or global warming/climate change?’ From this question, we obtain a dummy equal to one in case of ‘a fair amount’ or ‘a great deal’, and zero otherwise.
- ‘What do you think is the most important problem facing the country today?’ From this question, we obtain a dummy equal to one if the respondent answers ‘environment/pollution/climate change’, and zero otherwise.
- ‘Do you consider yourself an environmentalist or not?’ From this question, we obtain a dummy equal to one if the respondent answers ‘yes’, and zero otherwise.

As for CCES, data span the period 2006–2019. We employ the survey item on the preference scale between environmental protection and job availability. Specifically, we construct a dummy variable equal to one if the respondent thinks that environmental protection is ‘much more important’ or ‘somewhat more important’ than job availability, and zero otherwise. We also employ the survey item concerning support for the statement that states should be required to use a minimum amount of renewable fuels. Specifically, we construct a dummy variable equal to one if the respondent supports the statement, and zero otherwise (Supplementary Table 35). From the same data source, we also employ data on household income, employment status and union membership (Supplementary Table 35).

For European countries, we rely on Eurobarometer data, which span the period 2008–2019 and cover a total of 89,511 respondents in 13 countries (Switzerland and Norway are not included in the survey). We employ the following survey items:

- ‘How serious a problem do you think global warming/climate change is at this moment?’ This question is asked in every wave of the Special Survey on Climate Change and is akin to the Gallup one for the United States, thus allowing for a comparable analysis. Answers to this question range from 1 (‘not at all a serious problem’) to 10 (‘an extremely serious problem’). For comparability with the US analysis, we construct a dummy equal to one for scores greater or equal to 7, which is the sample mean.
- ‘To what extent do you agree or disagree with the following statement: fighting climate change and using energy more efficiently can boost the economy and jobs in the EU.’ From this question, we construct a dummy equal to one if the respondent ‘tends to agree’ or ‘totally agrees’ with the statement, and zero otherwise.

In addition to these two items, we employ a dummy equal to one if the respondent reports that their household income is ‘somewhat lower’ or ‘much lower’ than the very lowest net monthly income that their household would need to have a minimum acceptable standard of living, and zero otherwise. This item is only available for the year 2009. Moreover, we employ a dummy equal to one if the respondent expects the economic situation of their country to get worse over the next year, and zero otherwise. This item is only available for the years 2008 and 2011. Finally, we employ a dummy equal to one if the respondent ‘tends to agree’ or ‘totally agrees’ with the statement that protecting the environment should be a priority for their country, even if it affects economic growth. This item is only available for the year 2008.

Specifications

In this section, we outline all the estimated specifications.

Regional-level analysis on voting. For the regional-level analysis of voting in the United States, for example, in the first two columns of Table 1, we estimate specifications of the following form:

$$\text{Environmentalism}_{zt} = \beta \text{Trade Exposure}_{zt} + \alpha_z + \gamma_z \text{Trend}_{zt} + \varepsilon_{zt} \quad (4)$$

where the dependent variable is the environmentalism score of commuting zone z in election-year t . This is regressed on trade exposure computed at the commuting-zone level. α_z are commuting-zone fixed effects, while Trend_{zt} are commuting-zone-specific linear time trends, and ε_{zt} is an error term. Standard errors are clustered at the commuting-zone level.

For the regional-level analysis of voting in Europe, for example, in columns 3 and 4 of Table 1, we estimate specifications of the following form:

$$\text{Environmentalism}_{dt} = \beta \text{Trade Exposure}_{r(d)t} + \alpha_{r(d)} + \gamma_{r(d)} \text{Trend}_{r(d)t} + \varepsilon_{dt} \quad (5)$$

where d indexes electoral districts and t years. The dependent variable is the environmentalism score of district d in election-year t . The function $r()$ maps district d to its NUTS-2 region r , that is, the level at which we can measure the relevant trade exposure. In some cases, a district is itself a NUTS-2 region. In other cases, two or more districts are contained within the same region. Importantly, districts never span multiple regions. $\alpha_{r(d)}$ are region fixed effects, while $\text{Trend}_{r(d)t}$ are region-specific linear time trends, and ε_{dt} is an error term. Standard errors are clustered at the region-year level.

Individual-level analysis on voting and attitudes. The individual-level analysis for the United States, both in terms of voting and in terms of attitudes, is based on the following specification:

$$\text{Dep_Var}_{it} = \beta \text{Trade Exposure}_{z(i)t} + \mathbf{I}_{it}'\gamma' + \alpha_{z(i)} + \alpha_t + \varepsilon_{it} \quad (6)$$

where i indexes individuals and t election or interview years. The dependent variable is either the environmentalism score of the party voted by the individual in a given House election, or one of the survey items considered in the analysis of attitudes. In the vote analysis, the employed environmentalism scores are party-state specific, as in the commuting-zone level analysis. The function $z()$ maps individual i to their commuting zone of residence z , making it possible to attribute the relevant trade exposure value. $\mathbf{I}_{it}$ is a vector containing three individual controls: age, gender and a dummy for college education. α_z and α_t are commuting zone and year fixed effects, respectively, while ε_{it} is an error term. Standard errors are clustered at the commuting-zone-year level.

For the individual-level analysis for Europe, the estimated specification is:

$$\text{Dep_Var}_{it} = \beta \text{Trade Exposure}_{r(i)t} + \mathbf{I}_{it}'\gamma' + \alpha_{r(i)} + \delta_{r(i)} \text{Trend}_{r(i)t} + \varepsilon_{it} \quad (7)$$

where i indexes individuals and t years. $r()$ maps individual i to their region r . $\mathbf{I}_{it}$ contains individual controls for age, gender, and college education. $\alpha_{r(i)}$ are NUTS-2 region fixed effects, while $\text{Trend}_{r(i)t}$ are region-specific linear time trends, and ε_{it} is an error term. Standard errors are clustered at the region-year level. This specification is employed for the analysis of individual voting and for the analysis of the first two attitudes from the top of the plot in the right panel of Fig. 2, for which we have data spanning several years. For the other three survey items, which are available for only one or two years, the trends are dropped and the fixed effects are adjusted (see Supplementary Table 40 for details).

Robustness and extensions on the regional analysis

In the Supplementary Information, we report a large number of robustness checks and extensions to the regional analysis of voting. In particular, for both the United States and Europe, results are robust to: (1) excluding the years of the financial crisis (Supplementary Tables 5 and 20); (2) controlling for export exposure (Supplementary Tables 6 and 21); (3) controlling for ideological leaning in terms of protectionism

(Supplementary Tables 7 and 22); (4) controlling for macroeconomic conditions, that is, GDP growth and unemployment rate (Supplementary Tables 8 and 23); (5) controlling for exposure to extreme weather events, that is, temperature anomalies, heat episodes and dry spells, as in ref. 3 (Supplementary Tables 9 and 24); (6) controlling jointly for all the variables introduced at the above points (Supplementary Tables 10 and 25); and (7) controlling for potential confounders by interacting a number of presample political and economic characteristics of subnational geographic areas, both in levels and in changes, with either a linear time trend or year fixed effects, thus allowing for ex ante different areas to follow different trajectories over time (Supplementary Tables 11 and 26). Relatedly, in Supplementary Tables 12 and 27 we also provide evidence pointing to a lack of pre-trends. In addition, for the United States we also show that our evidence is robust to: (1) using alternative environmentalism scores based on party manifestos (Supplementary Table 13); (2) controlling for the vote share of the Democratic party in the first presample election, interacted with either a linear time trend or year fixed effects (Supplementary Table 14); and (3) considering both House and Senate elections at the level of analysis at which they are held: congressional-district level for the House, and state level for the Senate (Supplementary Tables 15 and 16). For Europe, we obtain similar results as the baseline when: (1) using as a dependent variable the cumulative share of district votes for parties identified as being green (Supplementary Table 28); (2) dropping the largest regions within each country and overall (Supplementary Table 29); (3) replicating the analysis at the more disaggregated NUTS-3 level (available for Italy and the United Kingdom; Supplementary Table 29); and (4) excluding countries with a majoritarian or mixed electoral system (Supplementary Table 30).

Robustness and extensions on the attitudes analysis

Several robustness checks and extensions to the analysis of attitudes are reported in the Supplementary Information, focusing on four main items of interest (that is, the two items on top of Fig. 2, for both the United States and Europe). In particular, results are largely robust when we consider exposure to imports from different origins (Supplementary Tables 36 and 41), and when we control for exposure to extreme weather events, as in ref. 3 (Supplementary Tables 37 and 42). In Supplementary Tables 33 and 34, and 38 and 39 we have performed the same heterogeneity analysis as for voting. The results show some variation as we consider different attitudes, along with some empirical regularities. For instance, the effect for women is never statistically different than for men. For one item in Europe, on the belief that fighting climate change can boost the economy, we see that none of the interactions is statistically significant, while the overall effect of import exposure is significant for all groups of respondents (Supplementary Table 39). This evidence points to a ‘sociotropic’ response of individuals, consistent with the results on voting. The picture remains largely sociotropic also for the other three items, although we find that some groups per each item are actually unaffected by import exposure. This is often (although not always) the case for students and young people, whose attitudes seem to be less related to the economic context. There are also some intriguing findings. For instance, when considering the European item on the seriousness of climate change (Supplementary Table 38), the overall effect of import competition seems to be driven mainly by respondents with high education and by white collars, while blue collars and low-education individuals do not seem to be significantly affected. This finding is again consistent with the role of sociotropic concerns, which are relevant also for social groups that are less likely to be directly negatively affected by trade exposure. At the same time, this evidence is also consistent with the idea that trade exposure may induce a shift in attitudes especially for workers who tend to display greener attitudes in the first place, thus potentially tilting public opinion at the margin in a way that is politically consequential.

Data availability

The data analysed in this study are available in the Harvard Dataverse repository at the following link: <https://doi.org/10.7910/DVN/T4ZAHS> ref. 25. All data are publicly available with one exception: individual data from the Gallup Poll Social Series. The Gallup-based replication database is shared upon request with researchers who have access to Gallup data.

Code availability

The data analysis was carried out in Stata and R. The codes that generate and visualize the results reported in this study are available in the Harvard Dataverse repository at the following link: <https://doi.org/10.7910/DVN/T4ZAHS> ref. 25.

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Author contributions

The initial research idea stems from I.C. and M.Z. C.B. was mainly responsible for preparing the election data for Europe, and the attitudes data for the United States and for Europe. M.Z. prepared the election data for the United States. I.C. and M.Z. prepared the trade exposure indicators. C.B., I.C. and M.Z. wrote the code and carried out the regression analysis. C.B. produced the figures. C.B., I.C. and M.Z. produced the tables. V.B. gave conceptual guidance and supported the theoretical framing and interpretation of the findings. I.C. wrote the original draft, and C.B., V.B. and M.Z. reviewed and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Charlotte Bez or Valentina Bosetti.

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